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A 650V Enhancement Mode GaN HEMT Device with Field Plate for Power Electronic Applications

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***Abstract*—GaN HEMTs are potential devices for future utilizations in power electronic applications. In this paper, an enhancement mode GaN HEMT power device is introduced. We present a schottky metal/p-type GaN gate to obtain positive threshold voltage and a source field plate to balance the electric field. Using Sentaurus TCAD as the simulation tool, the simulated breakdown voltages of the device with and without source field plate are 795V and 239V at L_{GD}=7μm, respectively. In order to obtain the highest breakdown voltage, the structure of the source field plate is analyzed and optimized before fabrication. The tested breakdown voltage of the fabricated device is 725V, and the tested threshold voltage is 1.15V.**

***Keywords*—GaN HEMT power device; p-GaN gate; field plate; breakdown voltage; threshold voltage**

I. Introduction

As the most representative wide band-gap semiconductor materials, GaN and SiC based devices are attracting global attention due to the ability of high power, temperature, frequency and anti-radiation [1, 2]. GaN based materials are better devices for achieving heterojunction structures, in which the narrow two-dimensional-electron-gas (2DEG) shows high electron mobility of 2000 cm^2/Vs, high saturation rate of 2.5E17 cm/s and high concentration of 10^{13} cm^{-2} [3-5].

In recent years, GaN HEMT MMICs have had greater utilizations compared with traditional GaAs MMICs in microwave field such as radar and communication systems [6]. Also, AlGaN/GaN heterojunction HEMT devices have great potential in power electronic field due to its high figure of merit [7], which may replace Si based devices in future. The critical breakdown voltage of GaN is 3.3 MV/cm, which is about 10 times higher than that of Si. This means that under similar breakdown voltage situation, the size of GaN device is way smaller than that of Si, which can effectively reduce that on-state resistance and improve the working frequency. Theoretically, GaN is more suitable for power device applications than Si or SiC as GaN has lower specific resistance ratio [8]. However, the existence of the 2DEG results in the phenomenon that the AlGaN/GaN HEMTs are normally-on even when the gate is unbiased, which is called the deletion mode. This means that the device is under the working mode even without gate drive device, which is a severe security threat for the system. Thus, normally-off AlGaN/GaN HEMTs are welcomed for the safety of circuit control. So far, extra fabrication processes such as recessed gate, fluorine ion implantation and P-GaN gate are widely used to form the enhancement mode AlGaN/GaN HEMTs [9], while each method faces their own difficulties in achieving high gate breakdown voltage and high threshold voltage.

In this paper, we proposed an enhancement mode GaN HEMT with a schottky metal/p-type GaN gate. Magnesium doped p-type GaN gate is designed to deplete the 2DEG under the gate, creating a positive threshold voltage V_{th}. Field plate techniques are used to adjust the electric field distribution and improve the breakdown voltage *BV*. Different lengths of field plate are simulated and analyzed to form the most optimized structure. And then, the device is fabricated on the basis of the best simulated results. Finally, the characteristics of the designed GaN HEMT and the purchased commercial superjunction MOSFET are tested and compared. The tested results indicate that even for the most advanced superjunction MOSFET of Si, the electrical parameters are still facing a great difference from the GaN HEMT.

II. Cell structure and mechanism

The designed cell structure is shown as in Fig. 1. The devices are simulated using Sentaurus TCAD software. From bottom to top, the cell consists of a 250 μm-thick Si substrate, a 2 μm-thick GaN buffer layer, a 20 nm-thick $Al_{0.2}Ga_{0.8}N$ barrier layer and a 100 nm-thick P-type GaN gate. The Si substrate can be replaced by SiC or sapphire substrate for better reliabilities, while Si is the most commercial and financial material. The interface between GaN and AlGaN forms a high density 2DEG due to polarization effects [10, 11]. Traditional AlGaN/GaN HEMT shows a negative threshold voltage due to the existence of 2DEG. However, positive threshold voltage device is more necessary as the turn-on and turn-off processes are under control.

The threshold voltage can be calculated as followed [12]:

$$V_{TH} = \frac{E_{f0}}{e} + \frac{\phi_B}{e} + \frac{\Delta E_C}{e} - \frac{d\sigma}{\varepsilon_i} - \frac{edN_{st}}{\varepsilon_i} - \frac{eN_b}{C_b} + \frac{e}{\varepsilon_i}\int_0^d dx \int_0^x (N_F(x))dx \quad (1)$$

Where Φ_B is the height of the schottky gate contact barrier, d is the thickness of the AlGaN barrier, ε_i is the dielectric constant, σ is the polarized charge density of the heterojunction interface, $\triangle E_C$ is the distance between two band gaps, E_{f0} is the distance between the Fermi energy level and the bottom of the conduction band, N_F, N_{st}, N_b are the ion doping density, interface trap density and the buffer layer trap density, respectively. Therefore, a magnesium ion doped p-type GaN gate is designed to deplete the 2DEG under the gate, improve Φ_B and create a positive threshold voltage. The doping concentration of magnesium is 3E17 cm^{-3}. Fig. 2 shows the electron density distribution under the gate with/without p-GaN, indicating that the electron density of the GaN HEMT with p-GaN is almost 1/10 of the one without p-GaN. Fig. 3 shows the simulated threshold voltage with/without p-GaN, explaining that the simulated threshold voltage of the p-GaN enhancement mode HEMT is 1.0V, while the threshold voltage of the traditional depletion mode GaN is -1.4V. Other parameters are set as followed: the source length L_S is 3 µm, the gate length L_G is 1.5 µm, the drain length L_S is 3 µm, the length between source and gate L_{GS} is 1.5 µm, and the length between drain and gate L_{GD} is 7 µm. The source and gate electrodes are contacted to AlGaN layer, connecting to 2DEG by tunneling effect.

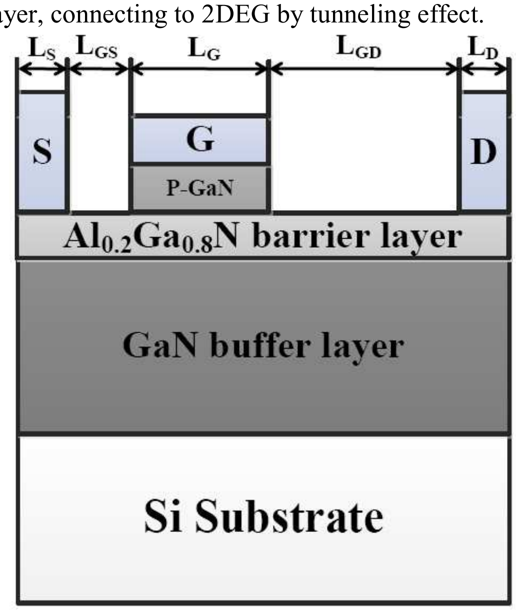

Fig. 1. The structure of the design cell.

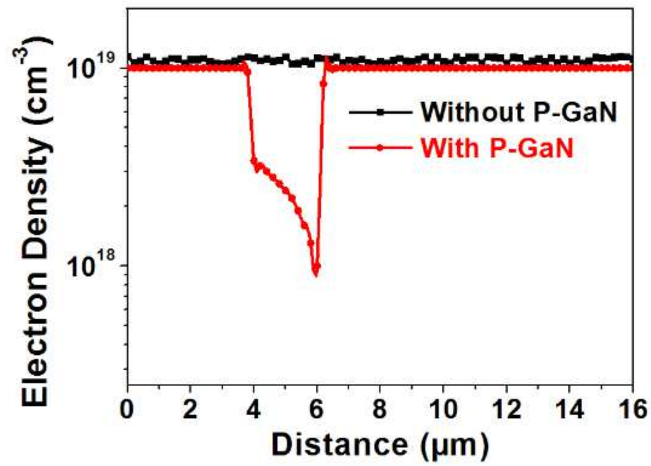

Fig. 2. The 2DEG density distribution under the gate with/without p-GaN.

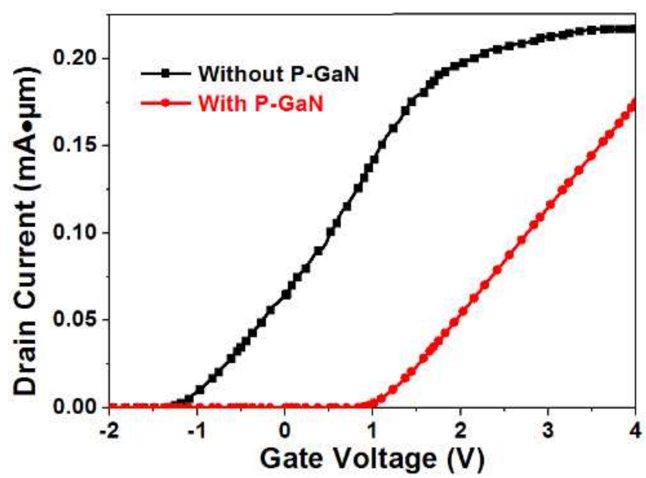

Fig. 3. The simulated threshold voltage with/without p-GaN.

Field plate structures are of great importance in GaN HEMTs as they can enhance the breakdown voltage by balancing the electric field distribution and decreasing the peak of the electric field near the gate edge towards the drain [13, 14]. Also, the field plate can reduce the current collapse effects [15, 16]. In the paper, source field plate and drain field plate are designed separately. Fig. 4 shows the schematic of the field gate. The lengths of the source field plate L_{SFP} and drain field plate L_{DFP} are simulated to determine the optimized size. The simulated breakdown voltage characteristic of the device with only source field plate is shown in Fig. 5. It indicates that the simulated breakdown voltage ranges from 410V to 795V when L_{SFP} ranges from 3 µm to 11 µm, while the simulated BV without field plate is 239V. The highest BV is 795V when L_{SFP} is 10 µm . Similarly, the simulated breakdown voltage characteristic of the device with only drain field plate is shown in Fig. 6. It is observed that the BV decreases as the drain field plate length L_{DFP} increases, where the highest BV is 205V when L_{DFP} is 4µm, even lower than the device without any field plate.

To explain the breakdown phenomenon, we analyzed the electric field along the 2DEG at a bias of V_{DS}=400V in Fig. 7. It indicates that for the device without field plate, the electric field peak of 4MV·cm^{-1} exceeds the GaN critical breakdown electric field of $3.3 MV \cdot cm^{-1}$. By contrast, for the device with source field plate L_{SFP}=10μm, there are two electric field peaks, while neither exceeds the GaN critical breakdown electric field. This shows that the field plate can effectively balance the electric field distribution, which enhances the breakdown voltage of the device. The field plate structure forms an extra electric field peak, which helps to disperse the electric field more uniformed. The fact that the electric field peak exists at the gate edge towards the drain suggests that the field plate are supposed to be designed near the source side, and this is the reason why source field plate plays a more important role in improving the *BV* than drain field plate.

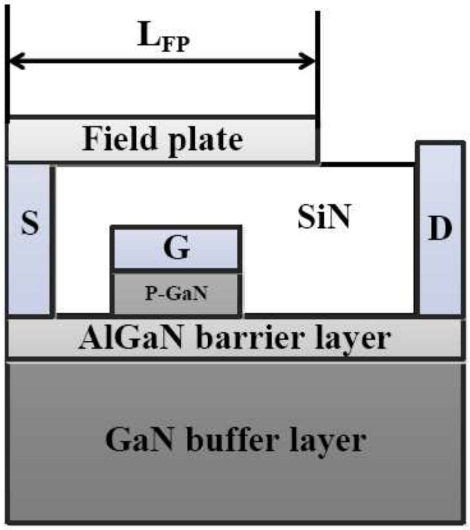

Fig. 4. Schematic of the field gate.

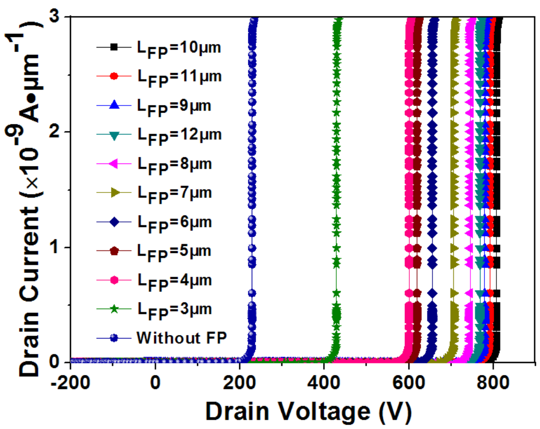

Fig. 5. The simulated breakdown voltage. L_{SFP} ranges from 3μm to 12μm. As a reference, device without field plate is plotted, as well.

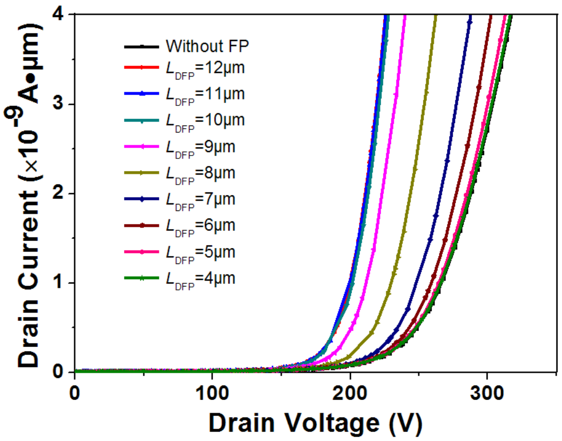

Fig.6. The simulated breakdown voltage of the cell with drain field plate. L_{DFP} ranges from 4μm to 12μm. As a reference, device without field plate is plotted, as well.

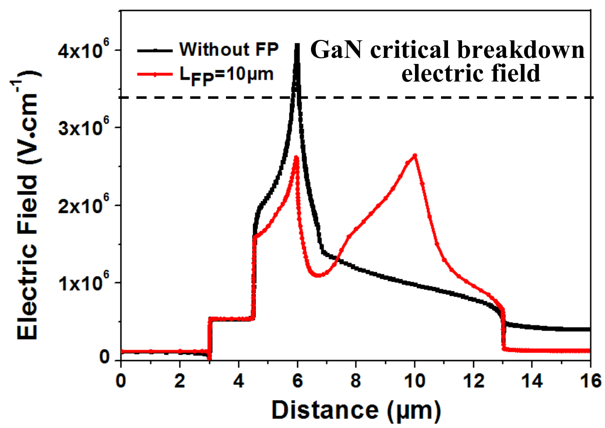

Fig. 7. The electric field along the 2DEG at a bias of V_{DS} =400V. The critical breakdown voltage of GaN is set to be 3.3 MV/cm.

Fig. 8 shows the equipotential lines of the device without field plate (a) and source field plate of 10μm (b) at their separate breakdown voltage. It is clear observed that the equipotential lines of the device with source field plate of 10μm is much uniformed than those of the device without field plate.

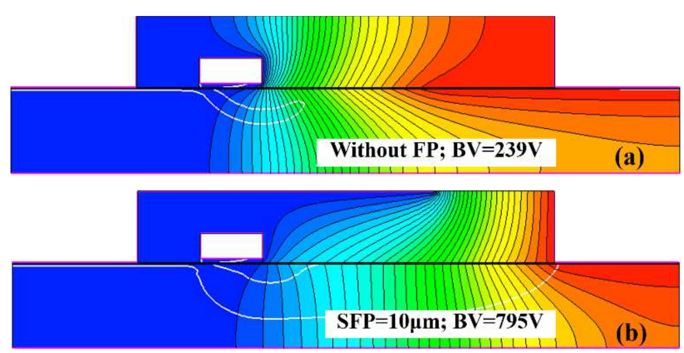

Fig. 8. Equipotential lines of the devices without field plate (a) at 239V and source field plate of 10μm (b) at 795V.

III. RESULTS AND COMPARISONS

For fabrication, the chip size is designed to be 2mm × 2mm, shown in Fig. 9. The device is fabricated on the basis of the simulated optimized cell structure with only source field plate of 10 μm. Table 1 summarizes the tested static and dynamic parameters. The tested BV is 725V, V_{TH} is 1.15V, and $R_{DS(ON)}$ is 0.23Ω. In particular, the reversed recovery charge Q_{rr} is 0, as no p/n junction exists in the heterojunction structure of AlGaN/GaN HEMT device. The test circuit of switching times and reverse channel characteristics are taken under the circumstance of T_A=25°C as shown in Fig. 10. The output characteristic is shown in Fig. 11. The largest output current is 12.8A. The tested $T_{d(on)}$, T_r, $T_{d(off)}$ and T_f indicate the ultrafast switching abilities of the GaN HEMT device for power application fields.

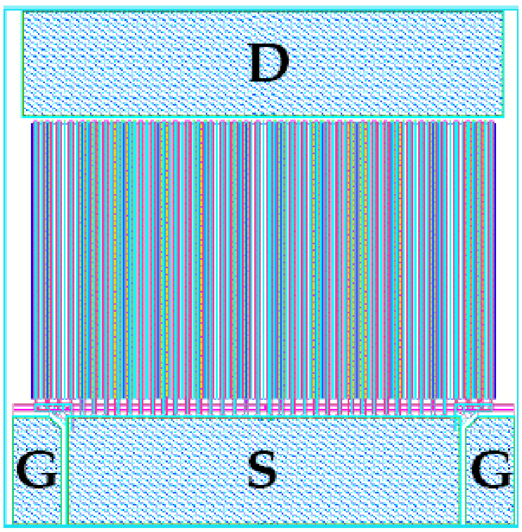

Fig. 9. Schematic of the chip. The chip size is 2mm × 2mm.

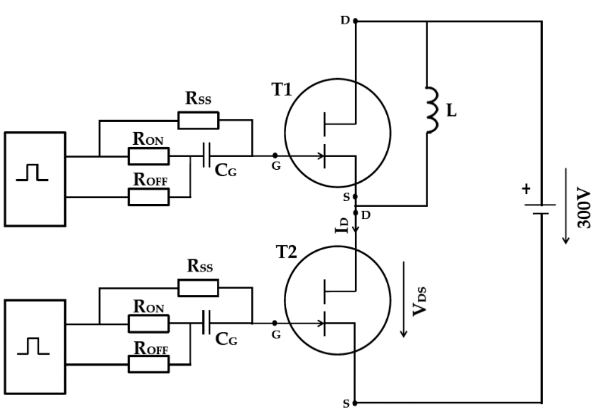

Fig. 10. The test circuit of switching times and reverse channel characteristics. I_D=1.5A, R_{ON}=10Ω, R_{OFF}=10Ω, R_{SS}=1kΩ, C_G=2nF.

TABLE I. TESTED PARAMETERS OF GAN HEMT DEVICE

Symbol	Test Condition (T_A=25℃)	Result	Unit
BV	V_{GS}=0V, I_D=0.5mA	725	V
I_{GSSF}	V_{GS}=5V , V_{DS}=0V	0.78	μA
I_{GSSR}	V_{GS}=-6V, V_{DS}=0V	0.01	μA
V_{TH}	V_{GS}=V_{DS}, I_D=1mA	1.15	V
$R_{DS(ON)}$	V_{DS}=5V, I_D=1.5A	0.23	Ω
C_{iss}	V_{GS}=0V, V_{DS}=300V, f=1MHz	62.5	pF
C_{oss}	V_{GS}=0V, V_{DS}=300V, f=1MHz	51.1	pF
C_{rss}	V_{GS}=0V, V_{DS}=300V, f=1MHz	8.3	pF
Q_G	V_{GS}=5V, V_{DS}=300V, I_D=1.5A	3.8	nC
Q_{oss}	V_{DS}=0 to 300V	18.5	nC
$T_{d(on)}$	See Fig. 9	2.9	ns
T_r	See Fig. 9	6.1	ns
$T_{d(off)}$	See Fig. 9	8.9	ns
T_f	See Fig. 9	2.7	ns
Q_{rr}	See Fig. 9	0	nC

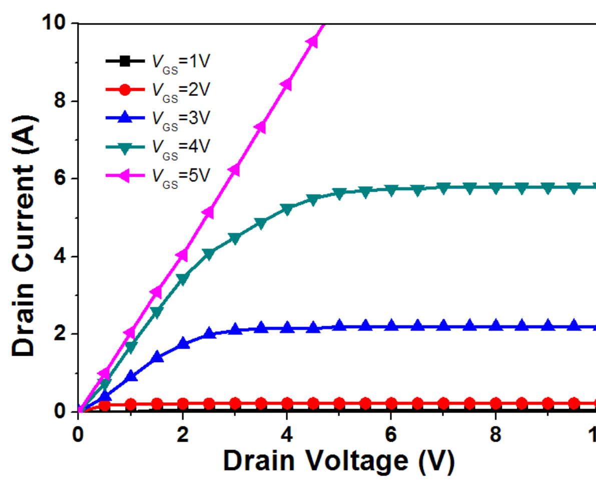

Fig. 11. The output characteristic of the device.

To better understand the dynamic behavior of the device, capacitances are in need of research. The gate-to-source capacitance (C_{GS}), gate-to-drain capacitance (C_{GD}) and drain – to-source capacitance (C_{DS}) between the electrodes are normally provided by the bias dependence of terminal capacitances, as followed. The tested capacitances are shown in Fig. 12.

$$C_{iss} = C_{GD} + C_{GS} \quad (2)$$

$$C_{oss} = C_{GD} + C_{DS} \quad (3)$$

$$C_{rss} = C_{GD} \quad (4)$$

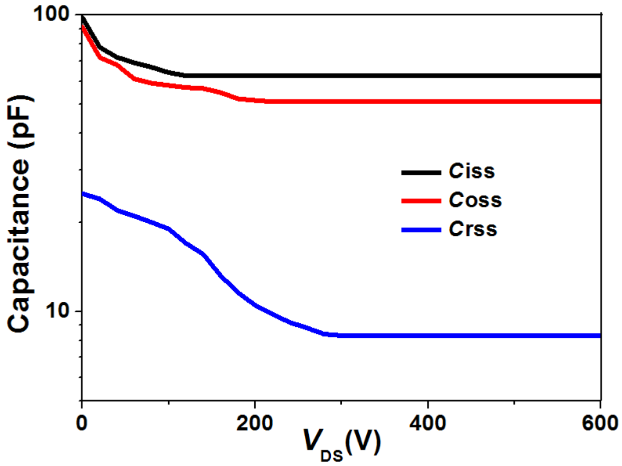

Fig. 12. The capacitances of the device.

For a deeper investigation of the excellent dynamic characteristics of the GaN device, we also tested the IPD65R225C7 from Infineon to compare the characteristic differences between the enhancement mode GaN HEMT and superjunction MOSFET. The parameters of IPD65R225C7 are listed as followed. The tested BV is 702V, V_{TH} is 3.45V, $R_{DS(ON)}$ is 0.225Ω, Q_G is 20.1nC, and C_{iss} is 996 pF. These results indicate that under similar BV and $R_{DS(ON)}$ situations, the Q_G and C_{iss} of the enhancement mode GaN HEMT are an order of magnitude smaller than those of the superjunction MOSFET. This means that compared with one of the most advanced Si devices, GaN devices still shows better electric characteristics, expressing an exceptional future in power device miniaturization and integration. Smaller gate charge and capacitance show that GaN HEMT can significantly reduce the switching loss in power electronic systems. However, the forward gate breakdown voltage of the enhancement mode GaN HEMT is only 5V, leading to the fact that specific gate drive is required when it is used in systems.

In future works, higher breakdown voltage is upon the request. Floating field plate and other complex field plate structures are gradually presented. On the other hand, gate breakdown voltage improvement is another engineering problems facing to be solved for the safety of the circuit systems.

IV. Conclusions

A 650V enhancement mode GaN HEMT device with field plate is proposed in this paper. The simulated BV, electron density and electric field of device with/without field plate are shown and analyzed, indicating that field plate can adjust the electric field distribution and significantly improve BV. The tested high BV of 725V, low resistance of 0.23Ω, ultrafast switching times of less than 10ns, low gate charge of 3.8nC and low capacitance imply that GaN HEMT has great potential for utilizations in power electronic fields. However, to achieve wider use in power systems, the enhancement mode GaN HEMT still faces the difficulties of achieving high gate breakdown voltage for the sake of circuit safety.

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